MISSION 1

The Questions

- Question 1

> Consider the two dimensional autonomous system :
>
> $$A:\begin{cases}\dot{x}=P(x,y)\\ \dot{y}=Q(x,y)\end{cases}$$
>
> Suppose our domain is simply connected , Is the following statment correct?
>
> $$divV=0 \iff A \text{ is a } Hamiltonian\ system$$

- Question 2

> Suppose we have the two dimensional autonomous system :
>
> $$A:\begin{cases}\dot{x}=P(x,y)\\ \dot{y}=Q(x,y)\end{cases}$$
>
> And the complex form of A is given by:
>
> $$\tilde{A}:\ \dot{z}=iz+Az^2+B\bar{z}^2+Cz\bar{z}$$
>
> Supoose :
>
> $$A:=A_1+iA_2$$
> $$B:=B_1+iB_2$$
> $$C:=C_1+iC_2$$
> $$\mu:=(A_1,A_2,B_1,B_2,C_1,C_2)$$
>
> Find the condition of μ such that $\tilde{A}$ is a >$Hamiltonian\ system.$

The Answers

- Answer of Question 1 :

> YES , it is correct.

PROOF:
$(\Leftarrow)$: We have the hypothesis that A is a Hamiltonian system That is $\exists\ H:\mathbb{R}^2\to\mathbb{R}\in C^2(\mathbb{R}^2)$ such that $\begin{cases}P(x,y)=H_y\\ Q(x,y)=-H_x\end{cases}$ Now , we have the definiton of $divV$:

> **Definition** : Suppose we have the two dimensional autonomous system :
>
> $$A:\begin{cases}\dot{x}=P(x,y)\\ \dot{y}=Q(x,y)\end{cases}$$

> Then we define the $divV := \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$

We want to show that $divV = 0$ So $divV \overset{by\ definition}{=} \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \overset{hypothesis}{=} H_{yx} + (-H_{xy}) \overset{H\ is\ in\ C^2}{=} 0 \square$
$(\Rightarrow)$:

> We need to introduce some definitions first :

> Definiton $1(1-form)$: We say that w is a $1-form$ if $w = A(x,y)dx + B(x,y)dy$ $where\ A, B$: $\mathbb{R}^2 \to \mathbb{R}$ are fucntions.

> Definiton $2(Exact\ form)$: Given a $1-form\ w = A(x,y)dx + B(x,y)dy$ we say w is $exact\ \text{ if }\ w = dH\ for\ some\ H$ Where $dH := \frac{\partial H}{\partial x}dx + \frac{\partial H}{\partial y}dy$

> Definiton $3(Closed\ form)$: Given a $1-form\ w = A(x,y)dx + B(x,y)dy$ we say that w is closed if $dw = 0$
> (where $dw := dA \wedge dx + dB \wedge dy = (A_x dx + A_y dy) \wedge dx + (B_x dx + B_y dy) \wedge dy = (B_x - A_y) dx \wedge dy$)

> Theorem$(Poincare\ lemma)$: Given a $1-form\ w := A(x,y)dx + B(x,y)dy$ and the domain of A, B are simply connected. Then w is $exact \iff w$ is $closed$

Now back to our proof, We have the hypothesis that $divV = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = 0$ We want to find H such that $H_y = P$ and $-H_x = Q$.

Since $dH = H_x dx + H_y dy = -Qdx + Pdy$

$$\text{So } \exists H \iff w := -Qdx + Pdy \text{ is } exact \tag{*}$$

Now compute dw :
$w := -Qdx + Pdy \Rightarrow dw = (P_x + Q_y)dx \wedge dy \overset{\text{hypothesis}}{=} 0dx \wedge dy = 0$
So w is $closed$
by $Poincare\ lemma$, we konw that w is $exact$, then by $(*)$ we konw that $\exists H$ That is , the system is $Hamiltonian.$ $\square$

- Answer of Question 2 :

> The condition is : $2A = -\bar{C}$ and B is free.

Consider the complex form of two dimensional autonomous system :

$$\dot{z} = iz + Az^2 + B\bar{z}^2 + Cz\bar{z} \tag{*}$$

where

$$A := A_1 + iA_2$$
$$B := B_1 + iB_2$$
$$C := C_1 + iC_2$$

Now we try to change this complex form into the original form Supoose

$$z := x + iy$$

Then $(*)$ is equal to

$$\dot{z} := \dot{x} + i\dot{y} = i(x+iy) + A(x+iy)^2 + B(x-iy)^2 + C(x+iy)(x-iy)$$

$\Rightarrow$

$\dot{z} = i(x+iy) + (A_1+iA_2)(x^2-y^2+2xyi) + (B_1+iB_2)(x^2-y^2-2xyi) + (C_1+iC_2)(x^2+y^2)$

$\Rightarrow$

$$\begin{aligned}\dot{z} \overset{\text{after simplify}}{=} & (-y + (A_1+B_1)(x^2-y^2) + (2B_2-2A_2)xy + C_1(x^2+y^2)) \\ & + i(x + (2A_1-2B_1)xy + (A_2+B_2)(x^2-y^2) + C_2(x^2+y^2))\end{aligned}$$

Now we have the original form :

$$\begin{cases}\dot{x} = -y + (A_1+B_1)(x^2-y^2) + (2B_2-2A_2)xy + C_1(x^2+y^2) \\ \dot{y} = x + (2A_1-2B_1)xy + (A_2+B_2)(x^2-y^2) + C_2(x^2+y^2)\end{cases}$$

Now compute the $divV = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} = (2(A_1+B_1)x + (2B_2-2A_2)y + 2C_1x) + ((2A_1 - 2B_1)x - 2(A_2+B_2)y + 2C_2y)$
In Question 1 , We konw that :

$$divV = 0 \iff A \text{ is a } Hamiltonian\ system$$

Let $divV = 0$,Then we have :

$$divV = (4A_1+2C_1)x + (-4A_2+2C_2)y = 0 \quad \forall x, y$$

So we have :

$$\begin{cases}4A_1 + 2C_1 = 0 \\ -4A_2 + 2C_2 = 0\end{cases}$$

$\Rightarrow$

$$\begin{cases}2A_1 = -C_1 \\ 2A_2 = C_2\end{cases}$$

That is :

$$\begin{gathered}2A = -\bar{C} \\ B \text{ is free}\end{gathered}$$

$\square$